When mutualism weakens: stress alters endophyte benefits in cool-season grasses

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Running header: Stress and Grass–Endophyte Interactions

Keywords: demography, mutualism, parasitism, plant–fungi interactions, range limits, stress gradient hypothesis

Submitted to: *Ecology letters* (Letter)

Data accessibility statement: If the paper is accepted, all data and computer scripts supporting the results will be archived in a Zenodo package, with the DOI included at the end of the article. During peer review, all data files required to reproduce the analyses, including annual census data, are temporarily hosted via Dropbox and available at `https://www.dropbox.com/scl/fo/fgxng2mcemjyv41bihul9/AGiZs5mPRGcCZURpGyvhjCE?rlkey=7mgdyrus3ga44yj3ygfncfmkd&dl=1`. All analysis scripts (Stan and R) are available at `https://github.com/jmoutouama/endo-range-limits`.

Conflict of interest statement: None.

Acknowledgments This research was supported by National Science Foundation Division of Environmental Biology award 2208857.

Authorship statement:

Abstract:
Main Text:
Figures: 5
Tables: 1
References: X

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Abstract

Benefits of plant-microbe symbioses are predicted to increase under environmental stress, but the relative roles of abiotic and biotic factors as modifiers of symbiotic interaction outcomes remain unclear. We studied how the independent and combined effects of aridity and herbivory modify the demographic effects of seed-transmitted fungal endophytes on three native grass host species using geographically distributed field experiments along a precipitation gradient, crossed with herbivore exclusion. Host fitness declined under low precipitation, with endophyte-symbiotic plants having twice lower survival than non-symbiotic plants, and spikelet production was frequently reduced under such stressful conditions. Herbivory had only occasional effects on plant–microbe symbioses. In contrast, abiotic stress emerged as the primary driver of mutualism outcomes, with effects opposite to those predicted by theory and previous studies. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

Introduction

Symbioses are widespread and ecologically important, occurring across both plants and animals, and often mediating key ecological functions such as stress tolerance and defense against natural enemies. These interactions are famously context-dependent, with the direction and strength of outcomes shaped by the environment in which they occur (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015). According to the stress-gradient hypothesis (SGH), positive interactions are generally expected to increase under stressful conditions, but actual outcomes depend on the type and intensity of biotic and abiotic stress (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015). For example, microbial symbionts that benefit host plants under drought stress may become neutral or even parasitic under more benign conditions (Giauque et al., 2019).

Because microbial symbioses are expected to become more beneficial in stressful conditions, they can substantially enhance host resilience to environmental stress. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation, or high temperature, symbiotic microbes can enhance host tolerance through a variety of mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including mycorrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation, allowing plants to maintain growth and metabolic function under drought or poor soil conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production of antioxidant enzymes, which reduce oxidative damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

While stress is hypothesized to strengthen the benefits of symbiotic microbes, growing evidence suggests that it could also elevate costs of symbiosis. Maintaining symbionts requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). For example, under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002)

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic effects of symbionts along environmental gradients. Most work has been conducted in controlled or common garden experiments. In *Ammophila breviligulata*, herbivory but not drought reduced growth, and no interaction was observed; however, plants hosting endophytic fungi were more resistant to these stressors (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function (Reynolds et al., 2003). Understanding how abiotic stress, biotic stress, and microbial symbiosis interact to influence plant performance requires experiments that jointly vary all three factors. However, to our knowledge, no such comprehensive study has been conducted. Here, we address this gap by examining the effects of biotic and abiotic stress on grass-endophyte symbioses in three grass species spanning a natural range-core/range-edge precipitation gradient.

Fungal endophytes provide a tractable system for studying how ecological context influences host–symbiont interactions, as host status (symbiotic vs. symbiont-free) can be experimentally manipulated. Grasses hosting endophytes often outperform endophyte-free individuals under water limitation, though the magnitude of these benefits depends on both host and symbiont identity (Decunta et al., 2021). A well-studied example is the association between cool-season grasses and seed-transmitted fungal endophytes in the genus Epichloë (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because vertical transmission links host and symbiont fitness, it is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). Environmental conditions, including long-term variation in precipitation and temperature, strongly influence both the prevalence of Epichloë infection and the benefits conferred to hosts (Fowler et al., 2025; Rudgers et al., 2016). Hosting Epichloë endophytes can entail costs, such as carbon allocation to sustain the fungus or risks associated with horizontal transmission; these may be offset by benefits including herbivore deterrence via fungal alkaloids (Clay et al., 2005; Saikkonen et al., 2010). Finally, Epichloë endophyte effects differ across host vital rates: infection may not improve survival under stress, but can still enhance growth and reproduction (Yule et al., 2013). Together, these features make grass-*Epichloë* symbioses an ideal system for investigating how abiotic stressors, such as drought, interact with biotic stressors, like herbivory, to determine the outcome of symbiosis.

Across an aridity gradient in the south-central United States, we investigated how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. This gradient, from mesic conditions in the east to increasingly arid conditions in the west, provides a biologically realistic stress gradient, and coincides with the westernmost range limits of many eastern North American grass species. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus excluding herbivores) to test its interaction with abiotic stress. We also examined the effects of Epichloë symbiosis by comparing endophyte-symbiotic and endophyte-free individuals across three species of native cool-season grasses.

We addressed how variation in precipitation and herbivory influences the outcome of grass-*Epichloë* symbioses. Specifically, we asked: (i) under reduced precipitation, do endophytes enhance host fitness by mitigating drought stress, or can they impose a net cost under extreme combined stress from drought and herbivory ? and (ii) does the relative importance of endophyte-mediated benefits change along the stress gradient, as predicted by the stress-gradient hypothesis (SGH), with higher abiotic stress amplifying mutualistic effects? We predicted that endophytes would enhance host fitness under moderate drought in the absence of herbivores, representing an endophyte-driven stress amelioration scenario (Fig. 1A). When herbivores are added as a second stress under more extreme drought, the metabolic cost of maintaining the endophyte may reduce benefits or even impose a net cost, corresponding to a reduced endophyte-mediated benefits scenario (Fig. 1B). Overall, consistent with the SGH, endophyte-mediated benefits are expected to increase under lower precipitation, with the presence or absence of herbivores modulating these effects.

Materials and methods

Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. *Agrostis hyemalis* hosts the endophyte

Epichloë amarillans (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë PauTG-1* or *E. typhina subsp. poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds and are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). Herbivory in this system is primarily from generalist insects and small mammalian grazers, which can influence both plant performance and endophyte-mediated defenses.

Source populations and symbiont removal

Plants used in our experiment were derived from natural (source) populations of each species (Fig. 2, Table S1). Prevalence of fungal endophytes in these natural populations ranged from XX% to XX% endophyte-positive hosts (NEW SUPP TABLE). [1] Seeds from these source populations (from ca. 30 plants per population) were either heat-treated or left untreated, intended to produce symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic lineages. For *Elymus virginicus* and *Poa autumnalis*, heat-treated seeds were placed in a drying over at 60°C for approximately five days. For *Agrostis hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. [2] All seeds, both heat-treated and non-heat-treated, were surface-sterilized with 10% bleach solution then germinated on agar plates. Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large in size they were vegetatively propagated, with clonal identities tracked across individuals. Once these plants were sufficiently large, they were vegetatively propagated, with clonal identities tracked across individuals. In total, we used 3, 5, and 4 populations (genotypes) for *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*, respectively, with 840, 840, and 720 clones in total across all populations. Before transfer to the field experiment, we confirmed endophyte status (E^{+} or E^{-}) of all seedlings using either microscopy or an immunoblot assay; realized numbers of E^{+} and E^{-} individuals per species and population are provided in Supplementary Table S3. For microscopy, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae (Jackson and Johnson-Cicalese, 1988). The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence (Koh et al., 2006). For

[1] *I am not sure about this.*

[2] *We should talk about this. You need more detail here – I have tried adding some but there is more to do. They were all treated but not all treatments were successful, which we need to report. And there were a few different methods used on different species. This should all go in the supplement. It is also not true that heat treatment has no effect on seed viability, which is why we tried to use seeds that were multiple generations removed from the heat treatment.*

genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applies equally to all clones of a given source genotype. Greenhouse-raised plants were planted into the field experiment in February-March 2023.

Common garden experimental design and climate data

To examine the effects of endophyte symbiosis across an abiotic stress gradient, we established common garden experiments with E+ and E- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S2). From west to east, mean annual precipitation increases more than three-fold across these sites, while mean annual temperature is relatively consistent (Fig. S1). For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 2A,B, C).

To characterize climate at these sites during the study period (2023–2025), we collected monthly temperature and precipitation data from PRISM at 800 m resolution (Hart and Bell, 2015), as daily data were not available for this period. For each site, we calculated mean temperature (°C) and cumulative precipitation (mm) over the corresponding census year, defined as the period from transplanting to the demographic census. During our study period, realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites (Fig. 2D).

At each site along the gradient, we established 24 square plots (1.5 m $\times$ 1.5 m), eight per host species. Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2E). The herbivory treatment targeted both vertebrate and invertebrate hebivores: exclusion plots had fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol, and access plots were fenced on two sides to control for any fence effect.

Each plot was planted with 15 individuals, a mix of E+ and E- host-plants. Initial endophyte prevalence varied from 20% (3 E+, 12 E-) to 80% (12 E+, 3 E-); this aspect of the experimental design was intended to address a different question about change in symbiont prevalence across generations, and is not further considered here. For each species at each site along the precipitation gradient, we planted a total of 60 E+ and 60 E- host-plants, half of those experiencing herbivore access and half experiencing herbivore exclusion.

The Kerrville and Sonora sites were destroyed and subsequently replanted in 2024. However, *Poa autumnalis* (POAU) was not replanted at the Sonora site.

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. Growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in years t and $t+1$. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds. Furthermore, to ensure that our herbivory treatment reflected herbivory experienced by individual plants. We defined damage as visible herbivore-induced foliar tissue loss and estimated the proportion of damaged plants per plot, site, and herbivore treatment.

Statistical modeling

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a zero-inflated negative binomial distribution, and spikelet count with a negative binomial distribution. All models included species-specific fixed effects for symbiosis (endophyte presence/absence), precipitation, and herbivory (herbivore access/exlusion), including all relevant two and three-way interactions. We considered models with linear and quadratic forms for the influence of precipitation, to account for the possibility of non-monotonic responses. Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S4). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

For each vital rate model, the expected value for the i^{th} observation (μ_i) was given by:

$$\begin{aligned}
\mu_i = &\beta_0[Sp_i] + \beta_P[Sp_i]precip_i + \beta_{P^2}[Sp_i]precip_i^2 + \beta_E[Sp_i]endo_i + \beta_H[Sp_i]herb_i \\
&+ \beta_{EP}[Sp_i]endo_i precip_i + \beta_{HP}[Sp_i]herb_i precip_i \\
&+ \beta_{HE}[Sp_i]herb_i endo_i + \beta_{HEP}[Sp_i]endo_i herb_i precip_i \\
&+ \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i} \\
&\qquad \phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\
&\qquad \rho_{\text{plot}[i],Sp_i} \sim N(0, \sigma^2_{plot}) \\
&\qquad \omega_{\text{source}[i],Sp_i} \sim N(0, \sigma^2_{source})
\end{aligned} \tag{1}$$

Here, $precip_i$ is total precipitation (mm) accumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence ($endo_i = 1$) or absence ($endo_i = 0$), and $herb_i$ indicates fencing treatment: partial access ($herb_i = 0$) or full exclusion ($herb_i = 1$). The model includes random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants. Random effect for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites to differ by species. This hierarchical structure allows the model to"borrow strength" across species (Gelman and Hill, 2007), improving estimates of fixed effects while still capturing species-specific responses to environmental gradient.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024). Models were run with 4 chains and sufficient iterations to ensure convergence, with weakly informative priors specified for all fixed effects. Full details on priors and model settings are provided in the accompanying code.

We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S2).

Drivers of endophyte effects on host demography

We used posterior parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each herbivory treatment (Access or Exclusion) at each level of precipitation, we computed posterior predictions of survival probability E^+ and E^- hosts using the Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits or a costs on vital rates under the given precipitation and herbivory conditions. When endophytes conferred a measurable cost, we additionally quantified the effect using a ratio-based metric (E^+ / E^-), where 1 indicates no effect, <1 indicates reduced performance of E^+, and >1 indicates enhanced performance. Both Δ and ratio metrics were summarized for low-climate (stress) conditions, defined as the lowest precipitation quartile, and interpreted only when posterior distributions indicated a significant difference between E^+ and E^- (i.e., 90% credible intervals excluding 0 for Δ or 1 for ratios), ensuring that reported costs or benefits reflect biologically meaningful effects.

To quantify statistical support for abiotic and/or biotic contexts as modifiers of grass-enodphyte symbiosis, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.1). All models included endophyte status as a baseline predictor but differed in whether they incorporated interactions with precipitation (Endophyte $\times$ Climate), herbivory (Endophyte $\times$ Herbivory), or both factors, including the three-way interaction (Endophyte $\times$ Climate $\times$ Herbivory). We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance. When the difference in ELPD between two models (ΔELPD) was small relative to its standard error (SE), we considered their predictive performance to be effectively similar. Accordingly, model rankings were interpreted cautiously, focusing on consistent trends across vital rates rather than on small differences between closely ranked models.

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Wetter conditions increased host survival across all species, and abiotic stress modified endophyte effects on survival for two of the three grasses. Grass-*Epichloë* symbiosis generally had negative or neutral effects under low precipitation (Table S5), but benefits increased under wetter conditions (Fig. 3; Fig. S3;Table S6). Moreover, our results confirmed that the herbivory treatment was effective, indicating that our experimental manipulations worked as intended (Fig. 2F).[3] Under herbivore access, *Agrostis hyemalis* and *Elymus virginicus* showed median $\Delta(E^{+}-E^{-})$ values from -0.07 to -0.01 at low precipitation ($\Pr(\Delta>0)=0.15$–0.42), increasing to 0.06–0.15 under high precipitation ($\Pr(\Delta>0)=0.78$–0.91). Herbivore exclusion weakened these effects: in *Agrostis*, Δ remained near zero across most precipitation levels, whereas *Elymus* shifted from strongly negative under dry conditions to positive under wetter conditions. *Poa autumnalis* showed little evidence of survival benefits under herbivore access ($\Delta=-0.05$ to 0.00; $\Pr(\Delta>0)=0.12$–0.53) and weakly positive effects under herbivore exclusion ($\Delta=0.00$–0.03), with credible intervals overlapping zero.

Endophyte presence influenced relative growth in all species, depending on herbivore access and precipitation (Fig. 4;Table S7). In *A. hyemalis*, growth effects were neutral to slightly negative under dry conditions and became increasingly positive at moderate-to-high precipitation when herbivores were present ($\Delta=0.27$–0.62; $\Pr(\Delta>0)=0.94$–0.99). Benefits under herbivore exclusion were smaller but consistently positive ($\Delta=0.08$–0.41; $\Pr(\Delta>0)=0.71$–0.96). *E. virginicus* exhibited weakly positive growth under herbivore access (Δ increasing modestly from dry to wet; $\Pr(\Delta>0)=0.52$–0.96) but small and uncertain effects under exclusion. In *P. autumnalis*, growth responses varied with precipitation and herbivore access: under herbivore access, Δ shifted from negative in dry conditions to positive in wet conditions, whereas under exclusion, benefits were strongest at low precipitation ($\Delta=0.56$–0.81; $\Pr(\Delta>0)=0.97$–0.98) but declined and became negative at the highest precipitation levels ($\Delta=-0.41$; $\Pr(\Delta>0)=0.03$).

Effects of *Epichloë* on reproduction were trait and species-specific. Inflorescence production showed little evidence of endophyte benefits in *A. hyemalis* or *E. virginicus*, with medians near zero and high uncertainty (Fig. 5;Table S8). By contrast, *P. autumnalis* exhibited strong positive effects, particularly under herbivore exclusion (Table S8). Spikelet production followed a similar pattern. *E. virginicus* and *P. autumnalis* had weak or negative effects under low precipitation, shifting to positive under wetter conditions, with benefits generally stronger when herbivores were excluded (Fig. S6; Fig. S7; Table S9). Credible intervals overlapped zero at low to moderate precipitation, indicating strong reproductive benefits were largely restricted to wetter conditions.

[3] *I would normally place this statement about the herbivory treatment's effectiveness in the Methods, since it's really confirming that our manipulations worked. However, since you requested to include it in the Results, I have done so here. The connection to the rest of the results isn't entirely seamless, but I hope this works.*

Abiotic factors as the primary drivers of fitness components across the stress gradient

The outcomes of grass-endophyte symbioses, whether beneficial, costly, or neutral, were primarily associated with climatic variation rather than herbivore treatment (Table 1). Across all vital rates (survival, growth, inflorescence, and spikelet production), model comparisons consistently indicated that models including an *Endophyte* $\times$ *Climate* interaction had equal or greater predictive support than alternatives emphasizing herbivory. Although differences in expected log predictive density (ELPD) were modest relative to their standard errors, these patterns were consistent across fitness components. For survival, the *Endophyte* $\times$ *Climate* model showed slightly higher predictive performance than the *Endophyte* $\times$ *Herbivory* model (ΔELPD$=-2.1$, SE = 2.8), whereas inclusion of the three-way *Endophyte* $\times$ *Climate* $\times$ *Herbivory* interaction reduced predictive performance (ΔELPD$=-5.0$, SE = 1.9).

Discussion

Drawing on three years of field experiments and Bayesian demographic models, we evaluated how biotic and abiotic stressors jointly influence the demographic effects of vertically transmitted Epichloë endophytes in three cool-season perennial grasses across a natural climatic gradient in the United States. Our results allow direct evaluation of the predictions outlined in our conceptual framework (Fig. 1). We predicted that endophytes would enhance host fitness under moderate drought in the absence of herbivores, representing an endophyte-driven stress amelioration scenario (Fig. 1A), and that the addition of herbivory under more extreme drought would reduce or reverse these benefits due to compounded metabolic costs (Fig. 1B). Contrary to these expectations, we found little evidence that endophytes buffered hosts against declining precipitation, even when herbivores were excluded. Instead, endophyte-mediated benefits were strongest under relatively benign conditions and weakened as abiotic stress intensified, while herbivory played a secondary and inconsistent role. These patterns do not support SGH-based predictions and instead suggest that increasing abiotic stress constrains both host performance and the capacity of endophytes to confer demographic benefits.

Endophyte presence often enhances host performance when abiotic resources, such as water, are sufficient to support both host and fungus, a pattern widely documented across grass-*Epichloë* systems (Clay, 1990). Endophytes can increase host growth, reproduction, and survival (Rodriguez et al., 2009), with benefits typically strongest under low abiotic stress, such as wetter conditions (Clay, 1988; Gibert et al., 2012), because hosts must invest carbon to maintain the symbiosis. Endophytes may also help hosts cope with environmental stress by promoting compounds that reduce oxidative damage during drought, pathogen attack, or other stressors (White Jr and Torres, 2010).

Although endophytes often confer benefits under favorable conditions, these advantages can decline as abiotic stress increases. Under low-precipitation conditions, reduced water availability may limit nutrient uptake and increase the metabolic burden of maintaining the endophyte, potentially tipping the cost-benefit balance against symbiotic plants (Ahlholm et al., 2002; Bronstein, 2001; Faeth, 2002). One possible mechanism is that severe drought can activate plant defense pathways, including elevated phytohormone production and reactive oxygen species, which may inhibit endophyte growth or reduce its functional contributions to the host (Bastías and Gundel, 2023; Eaton et al., 2011). Importantly, in our study system, mutualism weakening occurred even in the absence of herbivores, indicating that abiotic stress alone can disrupt the host-endophyte relationship. This contrasts with studies emphasizing herbivore-mediated context dependence (Afkhami and Rudgers, 2009; Bastias et al., 2017) and suggests that environmental limitations can independently generate similar outcomes.

Drought simultaneously restricts water and nutrient availability while increasing metabolic demands; thus, the host's carbon budget may be a key factor shaping the net costs and benefits of symbiosis. Water-stressed plants close stomata to conserve water, reducing photosynthetic carbon gain while still expending C to maintain cellular homeostasis (Dietze et al., 2014). Under these conditions, sustaining an endophyte may represent a relatively high carbon cost. Similar to other nutritional mutualisms, such as nitrogen-fixing rhizobia, water stress could reduce C allocation to the symbiont, although microbial partners may partially offset these costs by enhancing photosynthesis or water-use efficiency (Pringle, 2016; Toby Kiers et al., 2010). While carbon is a primary integrative currency, other factors including nutrient limitation and shifts in plant hormones also likely influence symbiosis outcomes. Future studies could quantify these costs by measuring nonstructural carbohydrates (NSCs) across tissues and over time along precipitation gradients, providing a clearer picture of how carbon limitation, in combination with other stressors, shapes the net outcome of endophyte-host interactions. Such approaches would improve understanding of the energetic balance of these interactions under varying abiotic conditions and help predict how environmental change may shift mutualism outcomes, with implications for both ecological theory and the management of plant-microbe systems.

Across the precipitation gradient, endophyte presence had little effect on overall inflorescence production, whereas spikelet production at the driest sites tended to be lower in endophyte-symbiotic plants. Although these effects were modest and uncertain, they suggest that drought-driven resource limitation may disproportionately constrain specific reproductive components. In low-resource environments, plants often delay reproduction, grow to a larger size before allocating energy to flowers and seeds, and may ultimately produce fewer reproductive structures (Wilson and Thompson, 1989). Consistent with this, spikelet production was more sensitive to resource limitation than inflorescence initiation, whereas overall vegetative growth remained unaffected by endophyte presence. This indicates that late-stage reproductive investment is constrained under high abiotic stress, even when vegetative growth is maintained. Such trait-specific sensitivity highlights the importance of examining multiple fitness components when evaluating symbiotic costs and benefits and aligns with demographic compensation, whereby reductions in one component may be partially offset by stability in others (Cheplick and Faeth, 2009).

Abiotic stress was the primary driver of endophyte-mediated effects across species and fitness components, while herbivory played a secondary role that depended on site-specific herbivore pressure. Water limitation strongly influenced host demography in our system, with drier sites constraining survival, growth, and fertility. Contrary to predictions of the stress gradient hypothesis (SGH), which posits that biotic interactions should become more positive under stressful conditions (Lortie and Callaway, 2006; Louthan et al., 2015), endophyte-mediated benefits were reduced under high abiotic stress. Most empirical tests of SGH have focused on plant–plant interactions (Bertness and Callaway, 1994), and while some recent syntheses suggest microbial interactions may show stronger support for SGH (Hammarlund and Harcombe, 2019; Piccardi et al., 2019), our results indicate that host-associated plant microbe mutualisms may not conform to this pattern. Unlike neighboring plants, which can reduce environmental stress for themselves and others for example, by shading soil, retaining moisture, or improving nutrient availability endophytes live entirely within their host and cannot directly modify the external environment. As a result, the benefits they provide depend on how well the host is coping with stress. When survival, growth, or reproduction is strongly limited by drought, the endophyte's capacity to enhance host fitness may be reduced. More broadly, our study suggests that outcomes of endophyte-mediated mutualisms are shaped not just by the presence of stress but by how stress constrains host performance, with important implications for how symbiotic responses may change under increasing drought associated with climate change.

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Figures

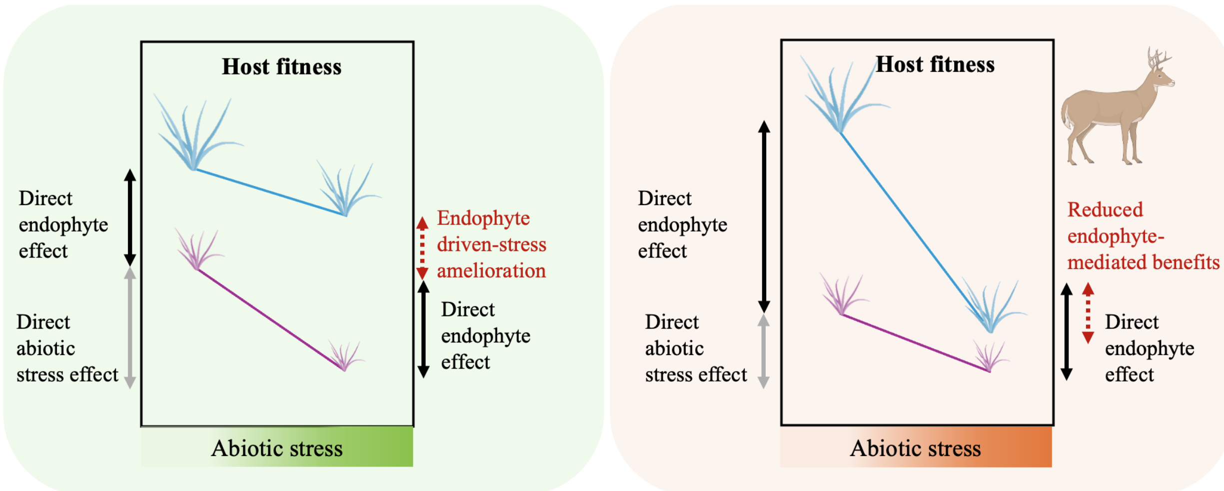

Figure 1: Conceptual framework illustrating the predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (a) **Endophyte-driven stress amelioration scenario:** Endophytes are predicted to enhance host performance under increasing abiotic stress (e.g., low precipitation) in the absence of herbivores by mitigating drought-related physiological costs. (b) **Reduced endophyte-mediated benefits scenario:** When abiotic stress is combined with herbivory, the positive effect of endophytes is expected to weaken. Limited plant resources are divided between defense against herbivores and sustaining the endophyte, reducing the endophyte's capacity to ameliorate drought stress and potentially resulting in a net cost of hosting an endophyte. Colors are used solely for visualization: blue lines represent endophyte-infected plants and violet lines represent endophyte-free plants.

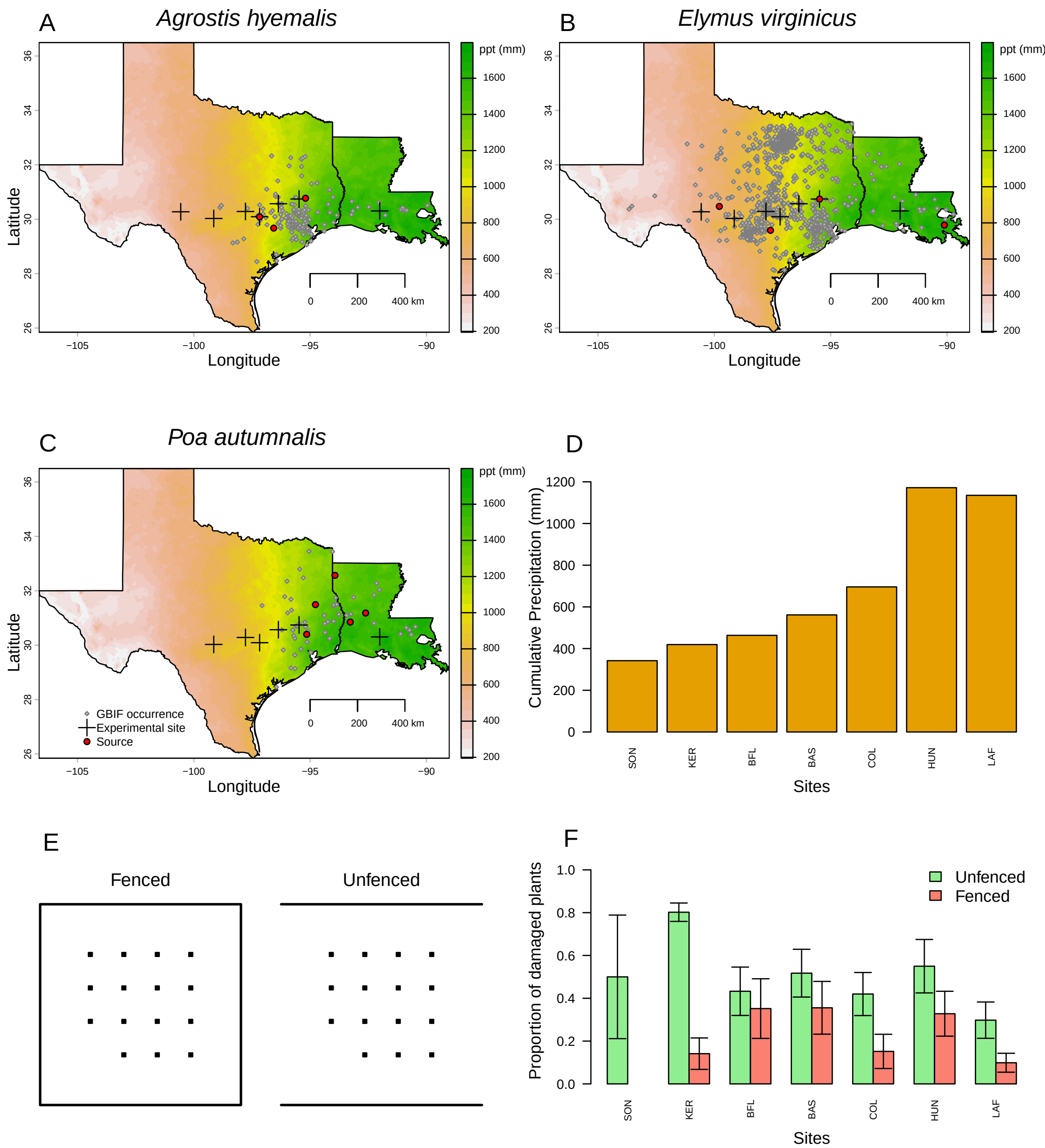

Figure 2: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1996–2025) from PRISM data (Hart and Bell, 2015), illustrating the precipitation gradient across experimental sites (scale on the right is in mm). Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). (D) Annual cumulative precipitation during each census year (May 2023 – June 2025) at each experimental site, ordered west to east. Values are also derived from PRISM climate data. (E) Schematic illustration of fenced (left) and unfenced (right) plots used to assess herbivory. Black squares represent individual plants; plot shape correspond to experimental design. (F) Proportion of damaged plants in fenced (salmon) and unfenced (light green) plots across experimental sites, with mean ± standard error. Sites are ordered west to east as in panel D. Panels E–F illustrate herbivory treatments and outcomes, and species-specific breakdowns are provided in Supplementary Figures.

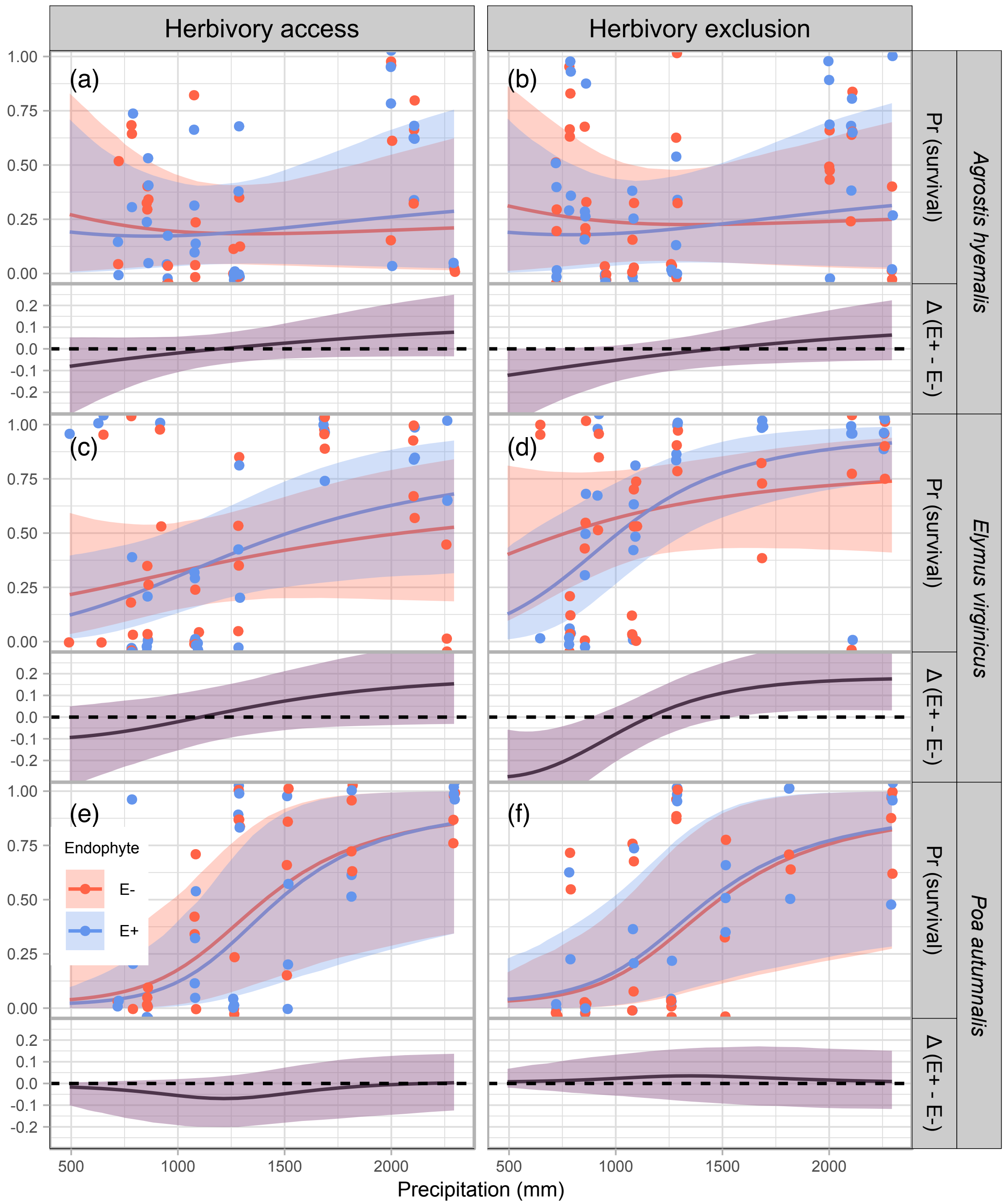

Figure 3: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

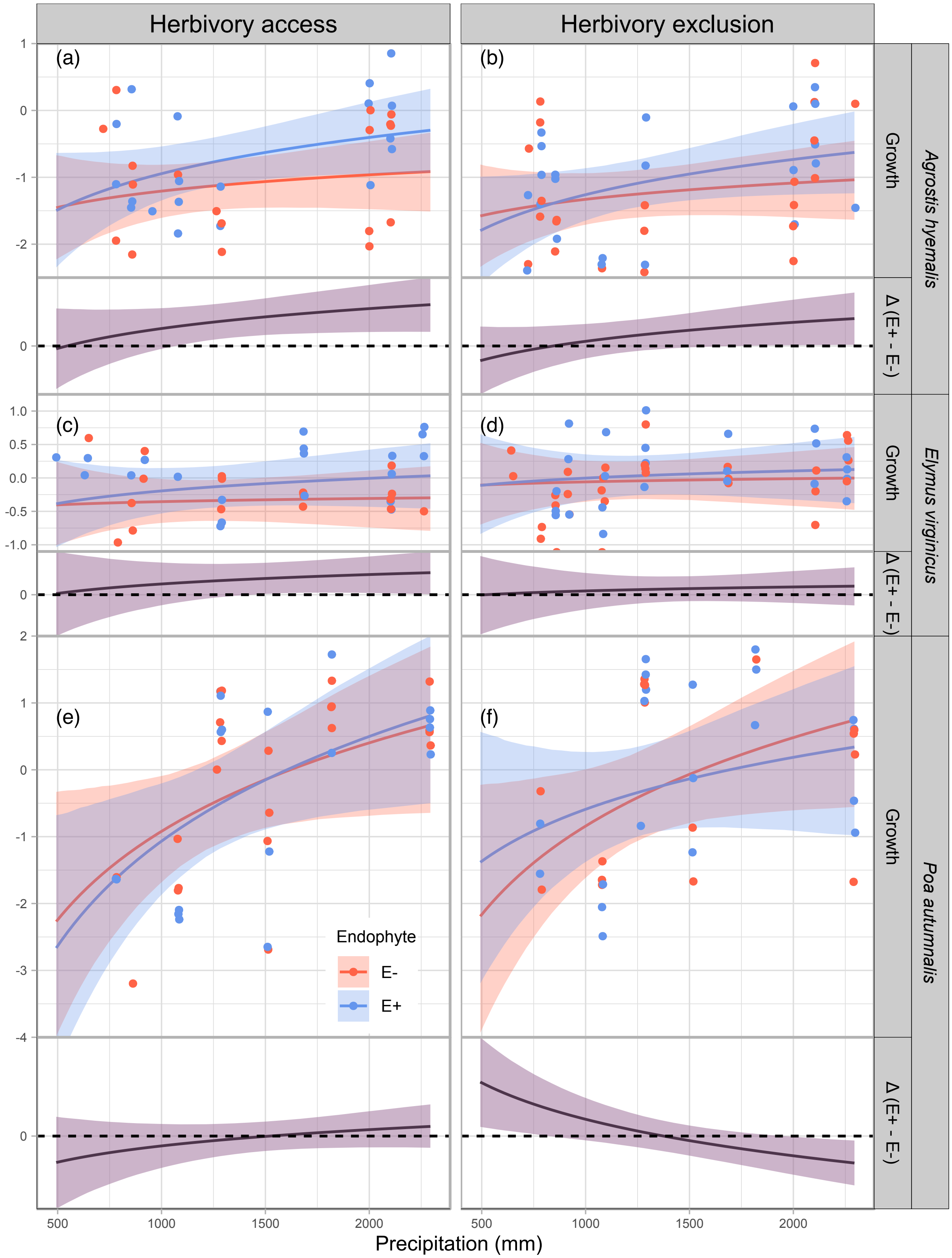

Figure 4: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

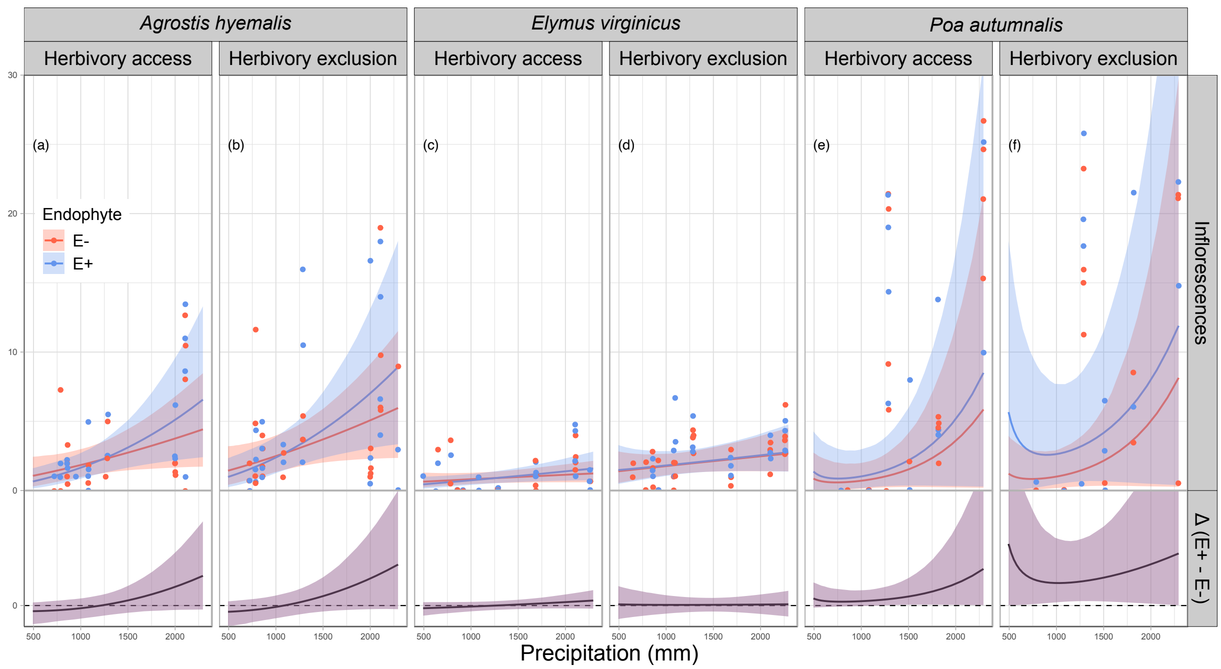

Figure 5: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+–E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means, jittered for visualization. Panels are grouped by species and herbivory treatment (Fenced, Unfenced).

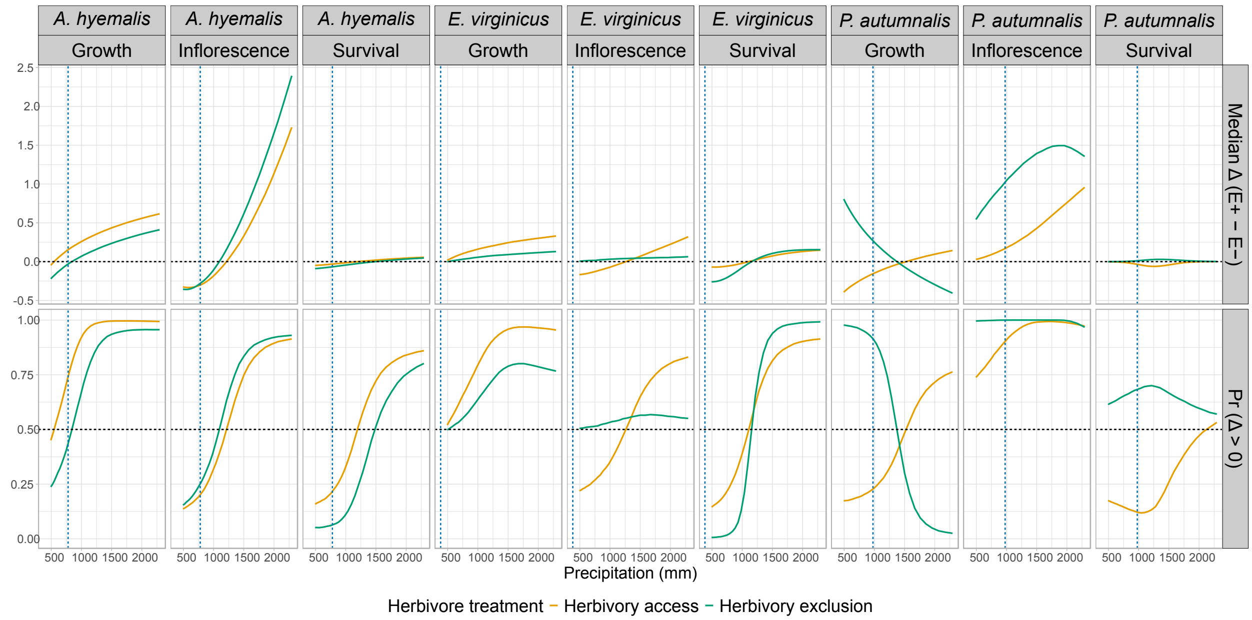

Figure 6: Predicted effects of herbivore exclusion on survival, growth, and inflorescence production across precipitation gradients for *A. hyemalis*, *E. virginicus*, and *P. autumnalis*. Panels show the median difference between fenced and unfenced treatments ($\Delta = E^{+} - E^{-}$) and the probability that this difference is positive ($\Pr[\Delta > 0]$). Horizontal dashed lines indicate $\Delta = 0$ and $\Pr(\Delta > 0) = 0.5$, and vertical dashed lines denote species-specific mean annual precipitation at range limits. Colors indicate herbivore exclusion treatments.

Tables

Table 1: Comparison of models assessing the drivers of grass–endophyte symbiosis across four fitness components (vital rates). ΔELPD represents the difference in expected log predictive density relative to the best-fitting model, and SE denotes the standard error of this difference.

Vital rate	**Model type**	**ΔELPD**	**SE**
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−4.65	2.05
	Endophyte × Climate × Herbivory	−5.23	3.82
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.27	3.24
	Endophyte × Climate × Herbivory	−0.31	3.51
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.74	2.70
	Endophyte × Climate × Herbivory	−0.90	3.39
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.92	3.31
	Endophyte × Climate × Herbivory	−3.27	2.44

S.1 Supporting Information

S.1.1 Supporting Methods

Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i + \beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i] endo_i herb_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i$$
$$+ \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i + \beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i] herb_i P_i + \beta_{\mathrm{herb}\times\mathrm{endo}}[\mathrm{Sp}_i] herb_i endo_i$$
$$+ \beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i] endo_i herb_i P_i + \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\mathrm{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\mathrm{endo}}[\mathrm{Sp}_i]$ and $\beta_{\mathrm{herb}}[\mathrm{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\mathrm{Sp}_i]$, while interactions are represented by $\beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{endo}\times\mathrm{herb}}[\mathrm{Sp}_i]$, and the three-way interaction $\beta_{\mathrm{endo}\times\mathrm{herb}\times P}[\mathrm{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{plot}})$.

S.1.2 Supporting Tables

Table S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	San Jacinto County, TX
AGHY	COHR	29.672833	-96.567517	Columbus, TX
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve[4]
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS[5]	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University Field Station, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S2: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S3: Realized numbers of endophyte-free (E−) and endophyte-infected (E+) individuals per species and population, with total number of clones.

Species	**Population**	**E−**	**E+**	**Total clones**
Agrostis hyemalis	COHR	238	242	480
Agrostis hyemalis	LPEF	227	236	463
Agrostis hyemalis	SJC	232	225	457
Elymus virginicus	HUNT	89	19	108
Elymus virginicus	JLP	156	135	291
Elymus virginicus	PALM	319	423	742
Elymus virginicus	RL5	141	81	222
Elymus virginicus	SHS	1	49	50
Poa autumnalis	LA1	119	101	220
Poa autumnalis	LA2	201	233	434
Poa autumnalis	LA7	89	72	161
Poa autumnalis	SFAEF	207	212	419

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.00
Survival	Linear	-0.055	1.09
Growth	Quadratic	0	0.00
Growth	Linear	-0.187	1.33
Inflorescence	Quadratic	0	0.00
Inflorescence	Linear	-1.510	1.43
Spikelet	Quadratic	0	0.00
Spikelet	Linear	-0.849	1.32

Table S5: Posterior summary of survival ratios (E^+ / E^-) under low-climate stress. Median, 90% credible intervals, and posterior probability that E^+ performs worse than E^- are shown for each species and herbivory treatment.

Species	Herbivory	Median ratio	Lower 90%	Upper 90%	Prob(E^+ worse)
Agrostis hyemalis	Herbivory access	0.666	0.253	1.36	0.825
Agrostis hyemalis	Herbivory exclusion	0.544	0.221	1.01	0.946
Elymus virginicus	Herbivory access	0.602	0.202	1.44	0.831
Elymus virginicus	Herbivory exclusion	0.352	0.0975	0.801	0.991
Poa autumnalis	Herbivory access	0.405	0.0785	1.83	0.837
Poa autumnalis	Herbivory exclusion	1.28	0.356	4.91	0.371

Table S6: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival across precipitation quartiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation bin (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Herbivory access	493	-0.049	-0.302	0.053	0.158
Agrostis hyemalis	Herbivory access	678	-0.038	-0.215	0.052	0.187
Agrostis hyemalis	Herbivory access	1036	-0.012	-0.099	0.058	0.357
Agrostis hyemalis	Herbivory access	1424	0.015	-0.050	0.102	0.681
Agrostis hyemalis	Herbivory access	2294	0.056	-0.035	0.249	0.860
Agrostis hyemalis	Herbivory exclusion	493	-0.090	-0.358	0.000	0.051
Agrostis hyemalis	Herbivory exclusion	678	-0.076	-0.264	0.002	0.057
Agrostis hyemalis	Herbivory exclusion	1036	-0.041	-0.142	0.019	0.127
Agrostis hyemalis	Herbivory exclusion	1424	-0.004	-0.085	0.072	0.448
Agrostis hyemalis	Herbivory exclusion	2294	0.047	-0.052	0.224	0.802
Elymus virginicus	Herbivory access	493	-0.069	-0.321	0.050	0.145
Elymus virginicus	Herbivory access	678	-0.065	-0.254	0.068	0.189
Elymus virginicus	Herbivory access	1036	-0.014	-0.143	0.113	0.424
Elymus virginicus	Herbivory access	1424	0.061	-0.069	0.194	0.779
Elymus virginicus	Herbivory access	2294	0.148	-0.031	0.356	0.914
Elymus virginicus	Herbivory exclusion	493	-0.259	-0.550	-0.058	0.006
Elymus virginicus	Herbivory exclusion	678	-0.230	-0.430	-0.058	0.012
Elymus virginicus	Herbivory exclusion	1036	-0.059	-0.183	0.061	0.210
Elymus virginicus	Herbivory exclusion	1424	0.090	-0.022	0.225	0.907
Elymus virginicus	Herbivory exclusion	2294	0.154	0.031	0.400	0.992
Poa autumnalis	Herbivory access	493	-0.001	-0.101	0.005	0.175
Poa autumnalis	Herbivory access	678	-0.005	-0.143	0.010	0.154
Poa autumnalis	Herbivory access	1036	-0.043	-0.196	0.022	0.118
Poa autumnalis	Herbivory access	1424	-0.051	-0.188	0.060	0.214
Poa autumnalis	Herbivory access	2294	0.000	-0.124	0.137	0.532
Poa autumnalis	Herbivory exclusion	493	0.000	-0.019	0.067	0.614
Poa autumnalis	Herbivory exclusion	678	0.001	-0.033	0.092	0.641
Poa autumnalis	Herbivory exclusion	1036	0.016	-0.058	0.136	0.689
Poa autumnalis	Herbivory exclusion	1424	0.029	-0.088	0.166	0.679
Poa autumnalis	Herbivory exclusion	2294	0.002	-0.117	0.151	0.570

Table S7: Posterior summaries of Δ ($E^{+} - E^{-}$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Herbivory access	493	-0.046	-0.643	0.561	0.450
Agrostis hyemalis	Herbivory access	678	0.092	-0.352	0.543	0.629
Agrostis hyemalis	Herbivory access	1036	0.274	-0.019	0.571	0.939
Agrostis hyemalis	Herbivory access	1424	0.412	0.147	0.683	0.994
Agrostis hyemalis	Herbivory access	2294	0.616	0.212	1.022	0.994
Agrostis hyemalis	Herbivory exclusion	493	-0.221	-0.723	0.290	0.237
Agrostis hyemalis	Herbivory exclusion	678	-0.090	-0.466	0.284	0.347
Agrostis hyemalis	Herbivory exclusion	1036	0.084	-0.165	0.336	0.710
Agrostis hyemalis	Herbivory exclusion	1424	0.214	-0.034	0.460	0.925
Agrostis hyemalis	Herbivory exclusion	2294	0.411	0.017	0.799	0.956
Elymus virginicus	Herbivory access	493	0.017	-0.618	0.647	0.520
Elymus virginicus	Herbivory access	678	0.084	-0.400	0.566	0.614
Elymus virginicus	Herbivory access	1036	0.168	-0.137	0.476	0.814
Elymus virginicus	Herbivory access	1424	0.232	-0.002	0.470	0.948
Elymus virginicus	Herbivory access	2294	0.330	0.008	0.643	0.955
Elymus virginicus	Herbivory exclusion	493	-0.002	-0.594	0.578	0.498
Elymus virginicus	Herbivory exclusion	678	0.023	-0.419	0.457	0.532
Elymus virginicus	Herbivory exclusion	1036	0.060	-0.205	0.316	0.651
Elymus virginicus	Herbivory exclusion	1424	0.087	-0.101	0.274	0.777
Elymus virginicus	Herbivory exclusion	2294	0.130	-0.160	0.410	0.767
Poa autumnalis	Herbivory access	493	-0.393	-1.089	0.289	0.174
Poa autumnalis	Herbivory access	678	-0.283	-0.803	0.230	0.185
Poa autumnalis	Herbivory access	1036	-0.136	-0.448	0.179	0.242
Poa autumnalis	Herbivory access	1424	-0.024	-0.245	0.202	0.431
Poa autumnalis	Herbivory access	2294	0.143	-0.175	0.474	0.763
Poa autumnalis	Herbivory exclusion	493	0.806	0.137	1.478	0.977
Poa autumnalis	Herbivory exclusion	678	0.556	0.058	1.060	0.967
Poa autumnalis	Herbivory exclusion	1036	0.223	-0.077	0.529	0.890
Poa autumnalis	Herbivory exclusion	1424	-0.028	-0.251	0.199	0.418
Poa autumnalis	Herbivory exclusion	2294	-0.406	-0.737	-0.067	0.025

Table S8: Posterior summaries of Δ ($E^{+} - E^{-}$) for flowering output at precipitation quantiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Species	Herbivory	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Herbivory access	493	-0.325	-1.371	0.229	0.137
Agrostis hyemalis	Herbivory access	678	-0.327	-1.230	0.320	0.172
Agrostis hyemalis	Herbivory access	1036	-0.157	-0.929	0.557	0.344
Agrostis hyemalis	Herbivory access	1424	0.231	-0.589	1.219	0.691
Agrostis hyemalis	Herbivory access	2294	1.730	-0.289	6.251	0.914
Agrostis hyemalis	Herbivory exclusion	493	-0.355	-1.550	0.311	0.154
Agrostis hyemalis	Herbivory exclusion	678	-0.332	-1.339	0.426	0.206
Agrostis hyemalis	Herbivory exclusion	1036	-0.067	-0.897	0.796	0.439
Agrostis hyemalis	Herbivory exclusion	1424	0.481	-0.492	1.772	0.800
Agrostis hyemalis	Herbivory exclusion	2294	2.394	-0.264	8.520	0.930
Elymus virginicus	Herbivory access	493	-0.167	-0.742	0.249	0.220
Elymus virginicus	Herbivory access	678	-0.142	-0.606	0.255	0.250
Elymus virginicus	Herbivory access	1036	-0.062	-0.410	0.274	0.369
Elymus virginicus	Herbivory access	1424	0.046	-0.277	0.391	0.606
Elymus virginicus	Herbivory access	2294	0.320	-0.226	1.141	0.831
Elymus virginicus	Herbivory exclusion	493	0.006	-1.001	1.434	0.505
Elymus virginicus	Herbivory exclusion	678	0.015	-0.824	1.125	0.512
Elymus virginicus	Herbivory exclusion	1036	0.033	-0.576	0.732	0.536
Elymus virginicus	Herbivory exclusion	1424	0.044	-0.445	0.566	0.561
Elymus virginicus	Herbivory exclusion	2294	0.064	-0.808	1.089	0.551
Poa autumnalis	Herbivory access	493	0.029	-0.152	1.709	0.738
Poa autumnalis	Herbivory access	678	0.070	-0.097	1.287	0.798
Poa autumnalis	Herbivory access	1036	0.194	-0.029	1.143	0.921
Poa autumnalis	Herbivory access	1424	0.385	0.054	1.767	0.987
Poa autumnalis	Herbivory access	2294	0.957	0.027	10.518	0.972
Poa autumnalis	Herbivory exclusion	493	0.543	0.021	14.821	0.996
Poa autumnalis	Herbivory exclusion	678	0.744	0.072	8.201	0.998
Poa autumnalis	Herbivory exclusion	1036	1.076	0.259	4.875	1.000
Poa autumnalis	Herbivory exclusion	1424	1.349	0.322	5.461	1.000
Poa autumnalis	Herbivory exclusion	2294	1.355	0.031	15.437	0.968

Table S9: Posterior summaries of Δ ($E^{+} - E^{-}$) at precipitation quantiles for each species and herbivore treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore treatment	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Elymus virginicus	Herbivory access	493	-3.149	-11.141	3.194	0.202
Elymus virginicus	Herbivory access	678	-2.375	-8.360	2.851	0.224
Elymus virginicus	Herbivory access	1036	-1.132	-5.122	2.668	0.303
Elymus virginicus	Herbivory access	1424	-0.095	-3.238	3.106	0.481
Elymus virginicus	Herbivory access	2294	1.676	-2.131	6.291	0.770
Elymus virginicus	Herbivory exclusion	493	-4.550	-11.872	1.236	0.092
Elymus virginicus	Herbivory exclusion	678	-3.364	-8.645	1.306	0.111
Elymus virginicus	Herbivory exclusion	1036	-1.444	-4.572	1.644	0.211
Elymus virginicus	Herbivory exclusion	1424	0.261	-2.038	2.652	0.577
Elymus virginicus	Herbivory exclusion	2294	3.229	-0.389	8.420	0.929
Poa autumnalis	Herbivory access	493	-0.584	-6.085	3.143	0.320
Poa autumnalis	Herbivory access	678	-0.584	-4.893	2.756	0.330
Poa autumnalis	Herbivory access	1036	-0.408	-3.240	2.226	0.373
Poa autumnalis	Herbivory access	1424	-0.025	-2.094	2.056	0.491
Poa autumnalis	Herbivory access	2294	1.097	-2.362	5.455	0.719
Poa autumnalis	Herbivory exclusion	493	2.630	-1.310	12.708	0.890
Poa autumnalis	Herbivory exclusion	678	2.943	-0.645	10.507	0.917
Poa autumnalis	Herbivory exclusion	1036	3.267	0.227	7.905	0.963
Poa autumnalis	Herbivory exclusion	1424	3.333	0.797	6.547	0.987
Poa autumnalis	Herbivory exclusion	2294	2.748	-1.344	8.544	0.871

Table S10: Posterior summary of spikelet ratios (E^+ / E^-) under low-climate stress. Median, 90% credible intervals, and posterior probability that E^+ outperforms E^- are shown for each species and herbivory treatment.

Species	Herbivory	Median ratio	Lower 90%	Upper 90%	Prob(E^+ worse)
Agrostis hyemalis	Herbivory access	0.835	0.566	1.21	0.785
Agrostis hyemalis	Herbivory exclusion	0.790	0.569	1.07	0.897
Elymus virginicus	Herbivory access	0.900	0.612	1.32	0.675
Elymus virginicus	Herbivory exclusion	1.35	0.928	1.98	0.0942

S.1.3 Supporting Figures

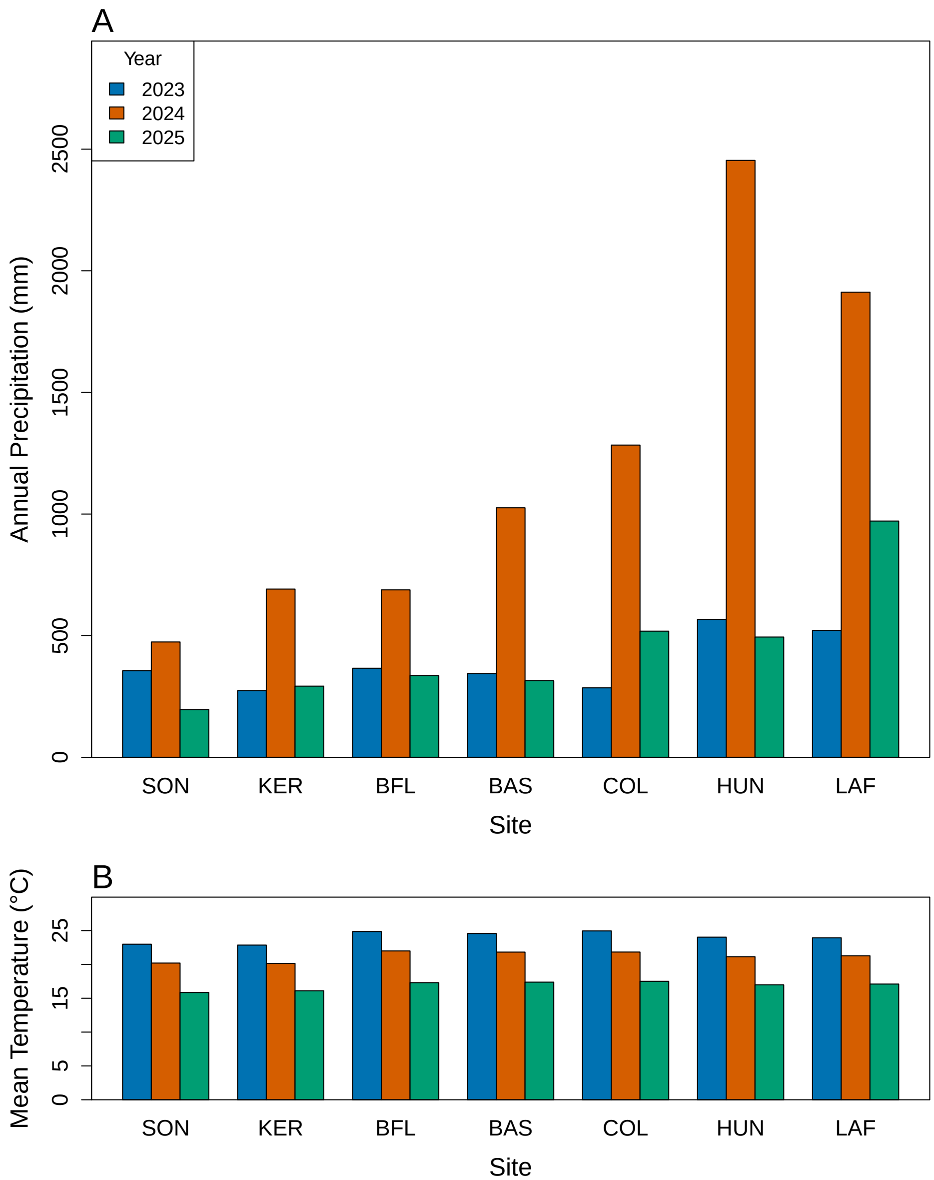

Fig S1: (A) Annual precipitation (mm) and (B) mean temperature (°C) for 2023–2025. Sites are arranged from west to east based on longitude.

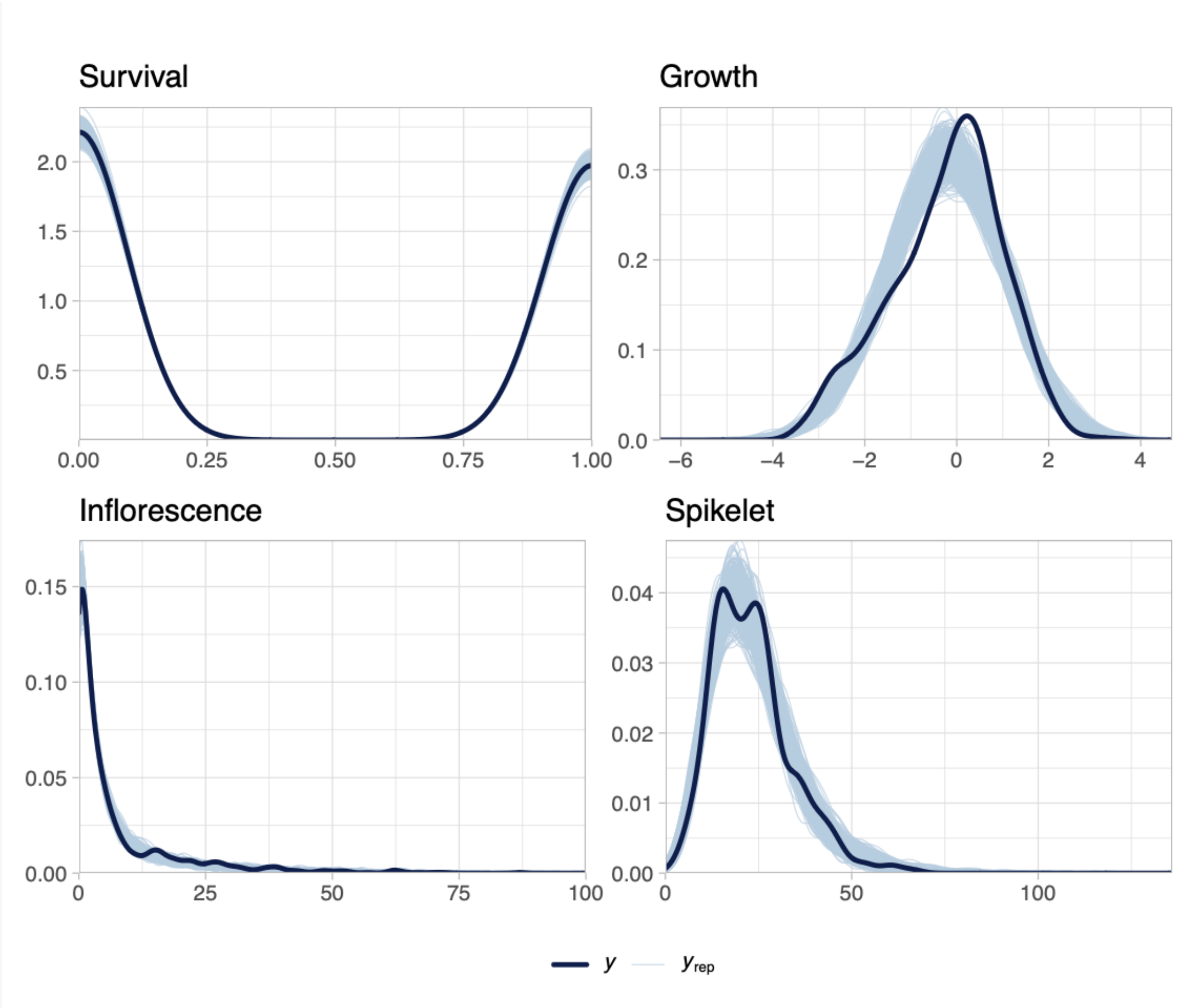

Fig S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

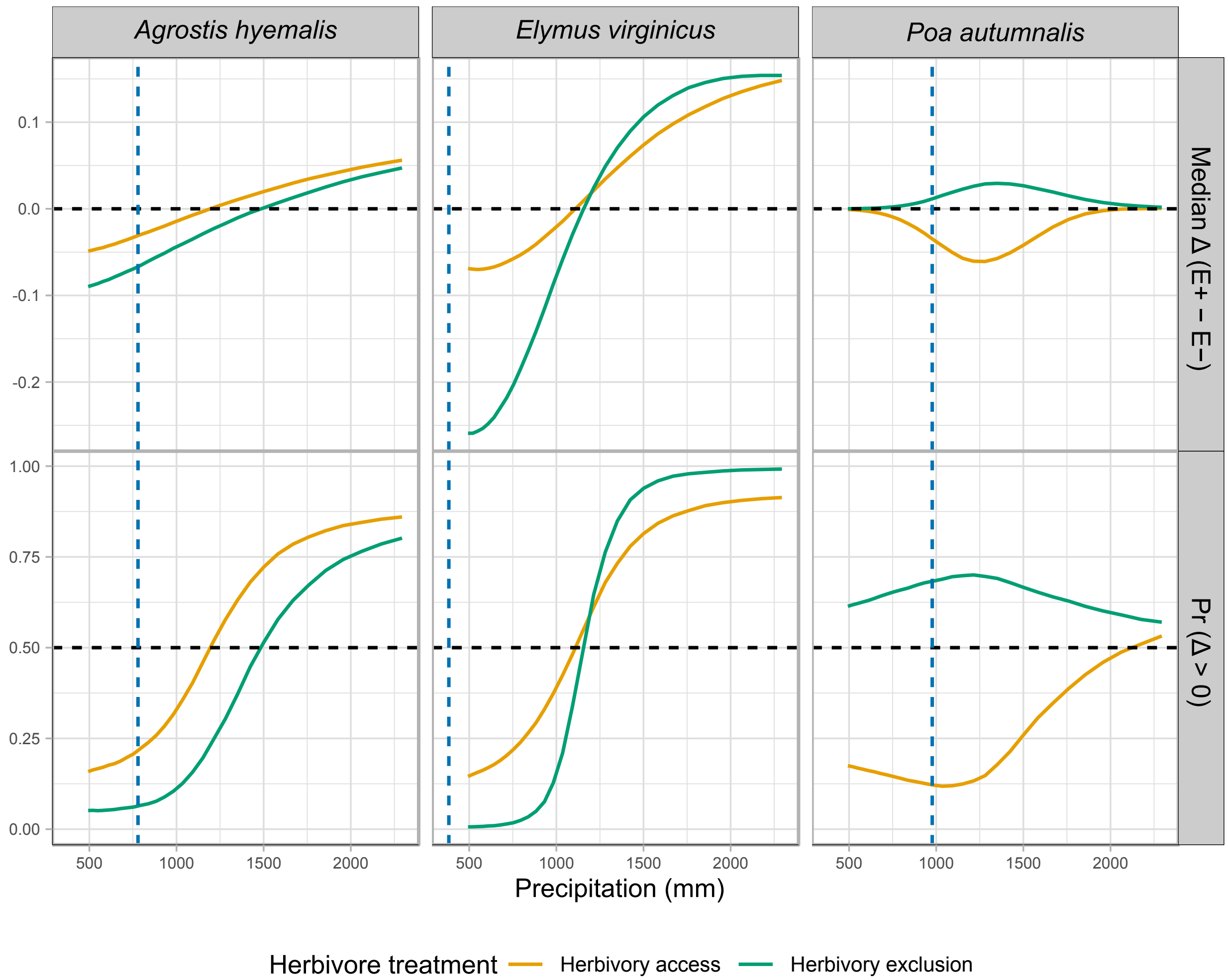

Fig S3: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

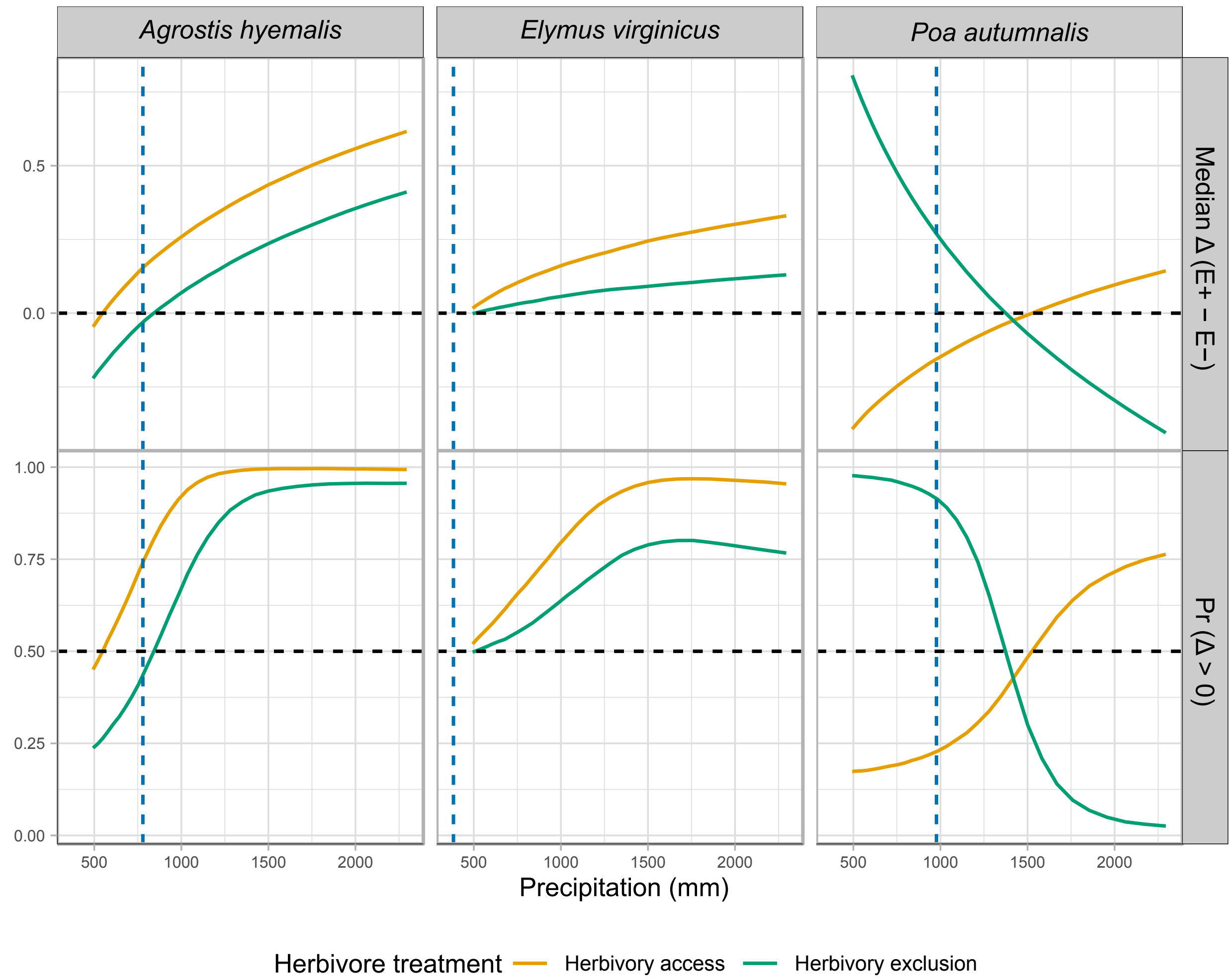

Fig S4: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^+) and endophyte-absent (E^-) plants (Median Δ, top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

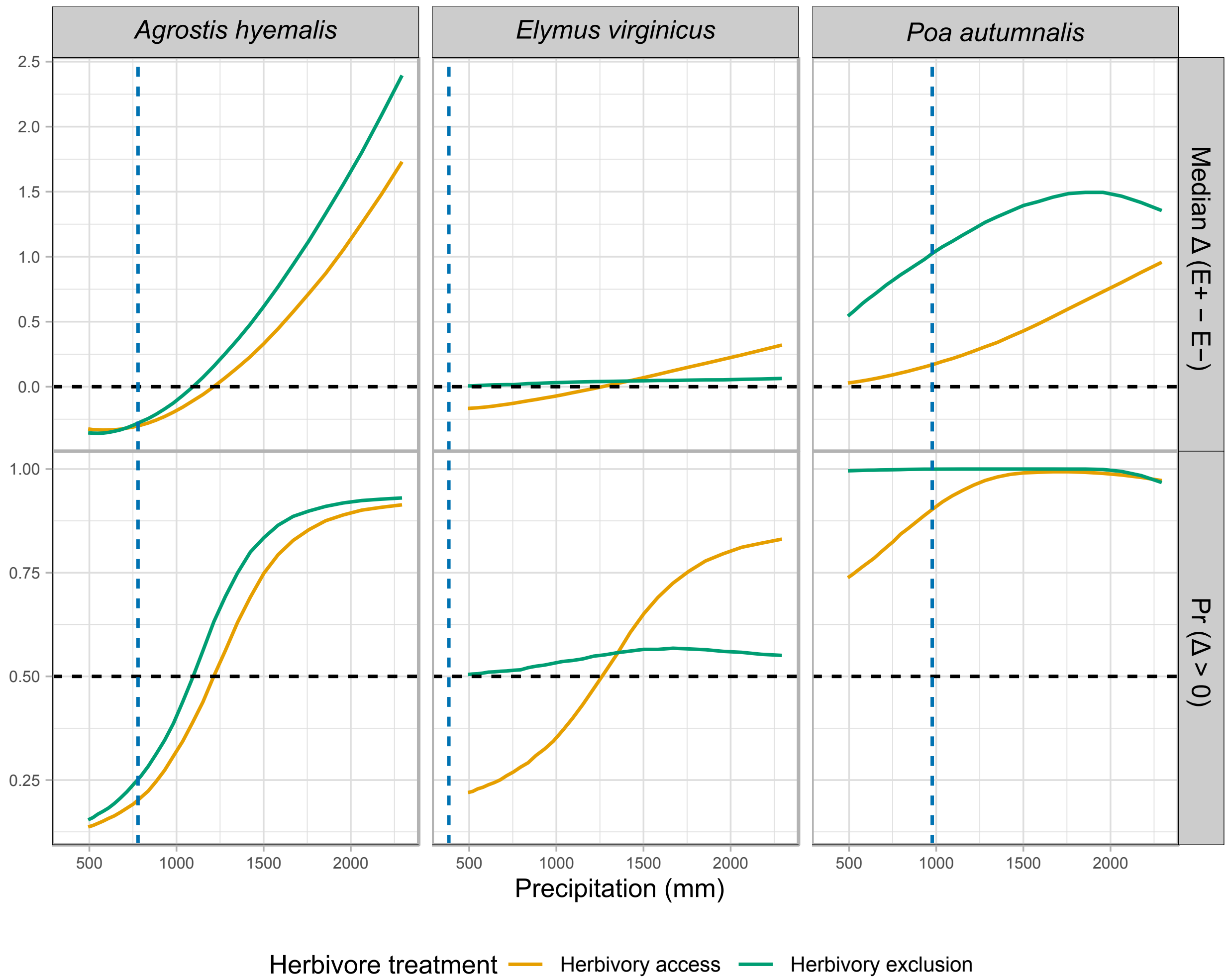

Fig S5: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ ($E^+ - E^-$) and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

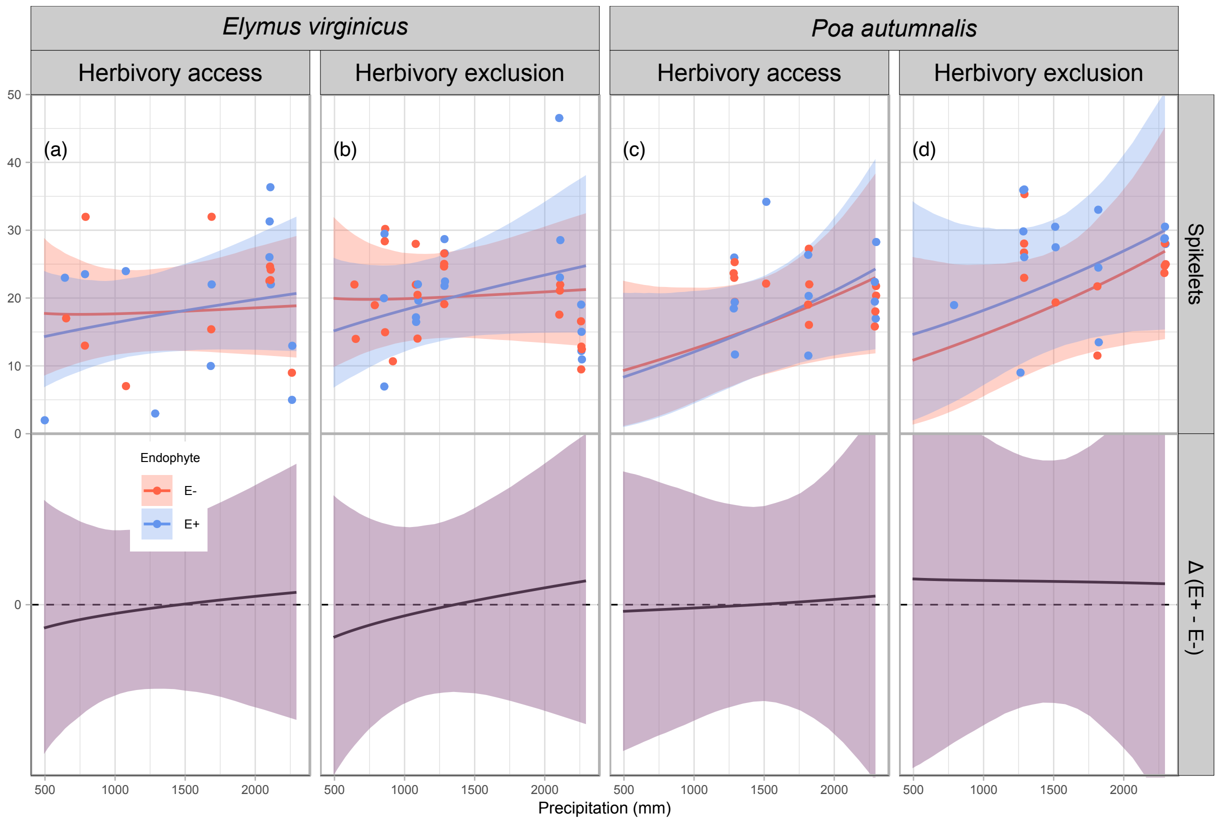

Fig S6: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

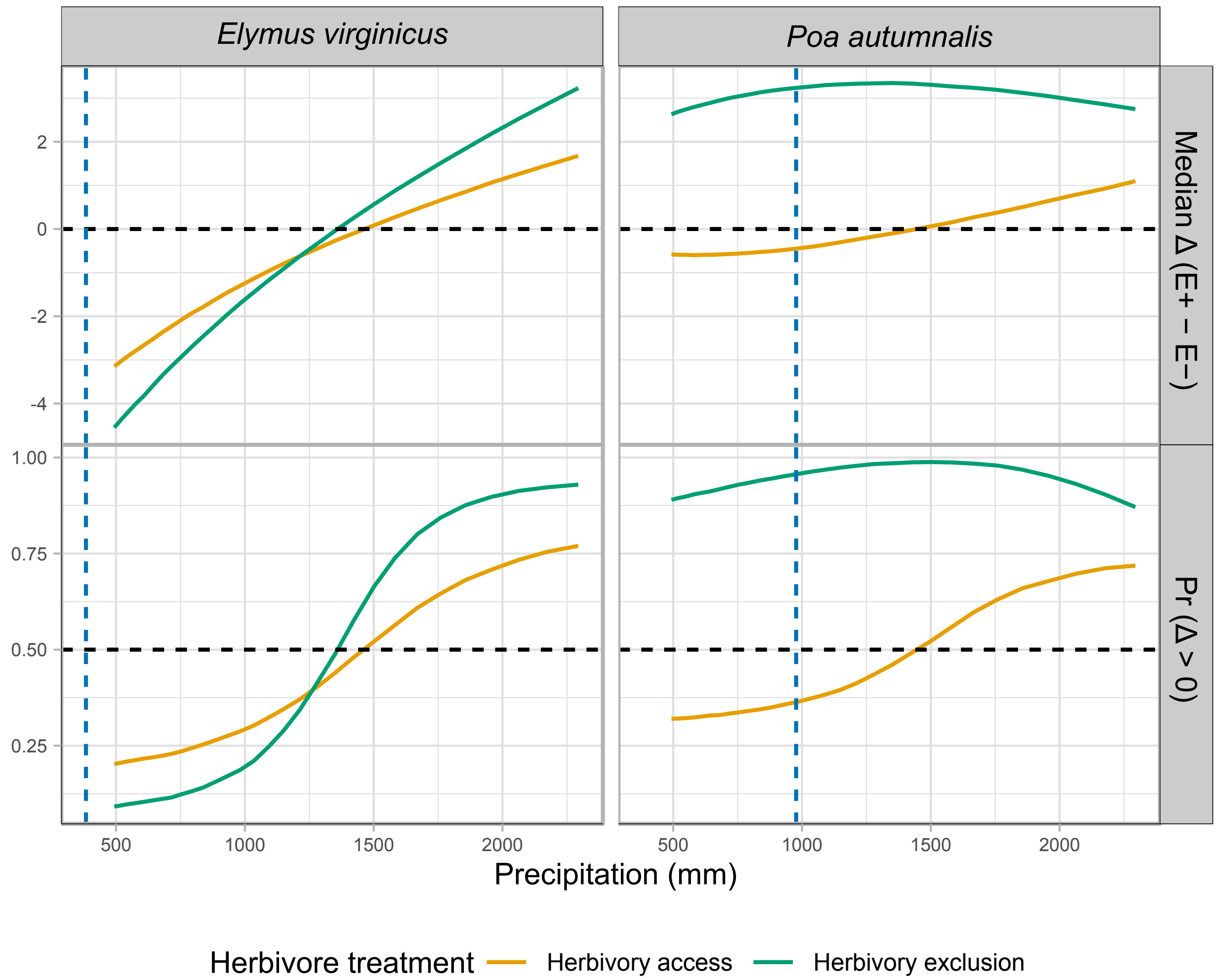

Fig S7: Effects of precipitation and endophyte presence on spikelet (reproductive output) for three cool-season grass species. Median Δ $(E^{+} - E^{-})$ and posterior probability $\Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $\Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.